

Student Performance Prediction Project Documentation

Overview

This project predicts student academic performance using machine learning techniques. The system classifies students as either **Pass** ($G3 \geq 10$) or **Fail** ($G3 < 10$) based on academic, demographic, and social factors.

The project combines:

- Data preprocessing
- Exploratory Data Analysis (EDA)
- Machine learning classification
- Model evaluation
- Streamlit live prediction dashboard

The goal is to identify academically at-risk students early and support data-driven educational interventions.

Dataset

The project uses the **UCI Student Performance Dataset** (student-mat.csv) developed by Paulo Cortez.

Dataset Information

- Source: Kaggle / UCI Machine Learning Repository
- Records: 395 students
- Features: 33 variables
- File Format: CSV (; separated)

Important Features

Feature	Description
G1	First period grade
G2	Second period grade
G3	Final grade
age	Student age
studytime	Weekly study time
failures	Number of previous failures
Medu	Mother's education
Fedu	Father's education

Feature	Description
traveltime	Travel time to school
internet	Internet access

Target Variable

The final grade (G3) was converted into a binary classification target:

Pass = $G3 \geq 10$

Fail = $G3 < 10$

Project Workflow

1. Data Loading and Exploration

The dataset was loaded into a Pandas DataFrame using:

```
pd.read_csv("student-mat.csv", delimiter=";")
```

Initial checks included:

- Missing value analysis
- Duplicate record detection
- Data type inspection
- Descriptive statistics

Findings

- No missing values
 - No duplicate rows
 - Dataset shape: 395 rows $\times$ 33 columns
-

2. Exploratory Data Analysis (EDA)

EDA was performed to understand relationships between features and student performance.

Key Analyses

- Distribution of final grades (G3)
- Correlation between G1, G2, and G3
- Effect of study time on grades
- Impact of previous failures

- Relationship between parental education and performance

Key Findings

- Strong positive correlation between G1, G2, and G3
 - Students with previous failures performed worse
 - Higher study time generally improved performance
 - Parent education positively influenced grades
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3. Feature Engineering and Preprocessing

Categorical variables were transformed into numerical format using one-hot encoding.

```
pd.get_dummies(df, drop_first=True)
```

Preprocessing Steps

- Encoded categorical variables
- Converted dataset into fully numeric format
- Prepared features for machine learning

The final encoded dataset contained 42 features.

4. Feature Selection

Correlation analysis identified the most important predictive features.

Important Predictors

- G1
 - G2
 - failures
 - Medu
 - studytime
-

5. Data Splitting

The dataset was divided into training and testing sets.

```
train_test_split(
```

```
X, y,  
test_size=0.2,  
random_state=42  
)
```

Split Ratio

- Training Set: 80%
 - Testing Set: 20%
-

6. Machine Learning Models

Two classification models were trained and evaluated.

Logistic Regression

Performance

- Accuracy: **92%**

Advantages

- Fast training
 - Strong classification performance
 - Easy interpretability
-

Random Forest Classifier

Performance

- Accuracy: **92%**

Advantages

- Handles nonlinear relationships
 - Robust against overfitting
 - Better feature interaction handling
-

7. Model Evaluation

Models were evaluated using:

- Accuracy Score
- Confusion Matrix
- Classification Report
- Precision
- Recall
- F1 Score

Both models achieved strong classification performance with comparable accuracy.

8. Model Export

Trained models were saved using joblib.

Exported Files

logistic_regression_model.joblib
random_forest_model.joblib

9. Streamlit Web Application

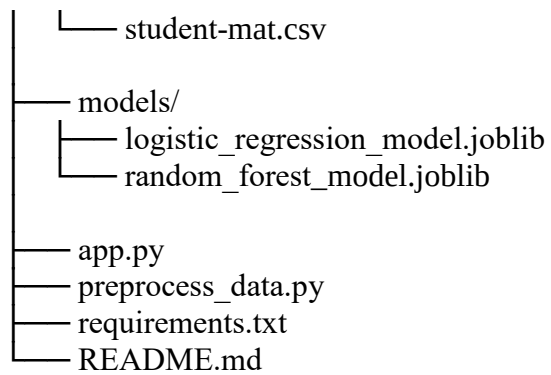
A Streamlit application (app.py) was developed for live student performance prediction.

Features

- Interactive prediction dashboard
 - User-friendly interface
 - Real-time pass/fail prediction
 - Educational risk assessment
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Project Files

```
Student-Performance-Prediction/  
├── notebooks/  
│   └── analysis.ipynb  
└── data/
```



Installation

Install required dependencies:

```
pip install -r requirements.txt
```

Run the Streamlit Application

```
streamlit run app.py
```

After running, the application will open in the browser at:

```
http://localhost:8501
```

Technologies Used

- Python
 - Pandas
 - NumPy
 - Scikit-learn
 - Matplotlib
 - Seaborn
 - Streamlit
 - Joblib
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Future Improvements

- Hyperparameter tuning
- Advanced ensemble models
- Feature importance analysis

- SHAP explainability integration
 - Docker deployment
 - Cloud deployment with Streamlit Community Cloud
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Conclusion

This project demonstrates how machine learning can be used to predict student academic performance and identify at-risk students early.

By combining predictive analytics, feature engineering, and interactive dashboards, the system provides a practical educational decision-support tool for improving student success outcomes.